Within−patient pairs

NF0014_T1 (n = 1225) | NF0014_T2 (n = 1225) | NF0018_T6 (n = 1225)

NF0021_T1 (n = 1225) | NF0035_T1 (n = 1225) | NF0037_T1 (n = 1770)

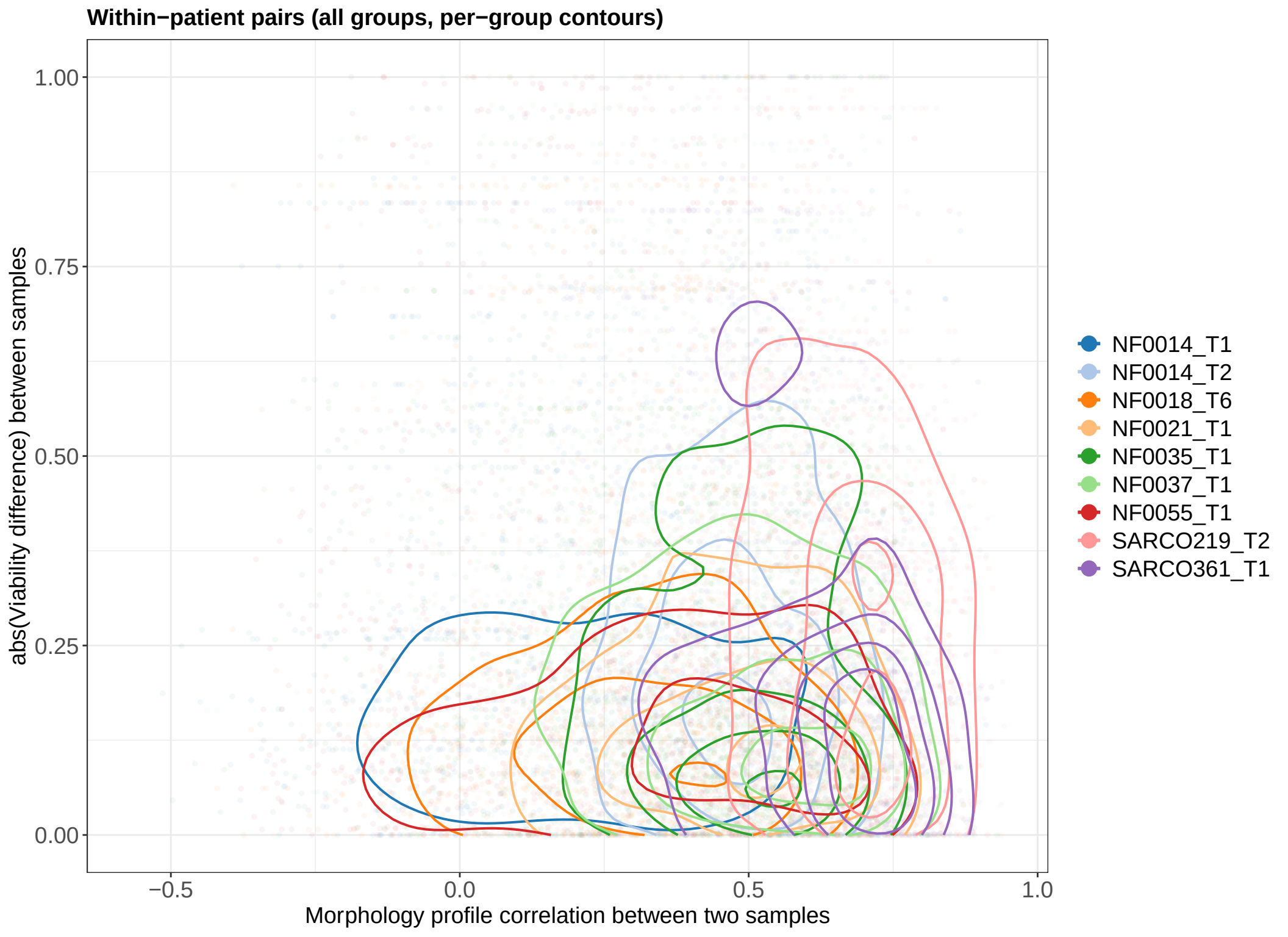
NF0055_T1 (n = 1378) | SARCO219_T2 (n = 1225) | SARCO361_T1 (n = 1225)

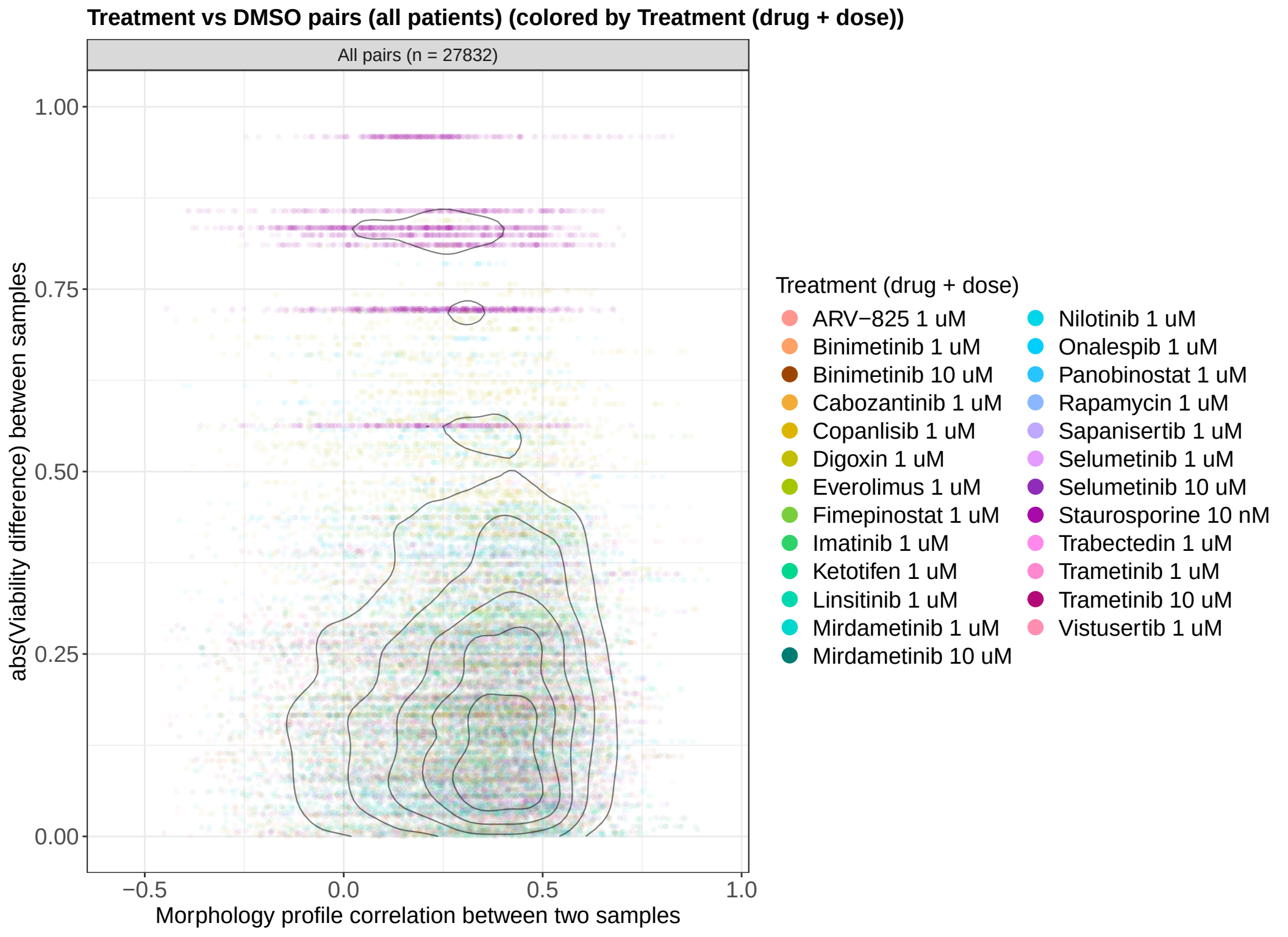
abs(Viability difference) between samples

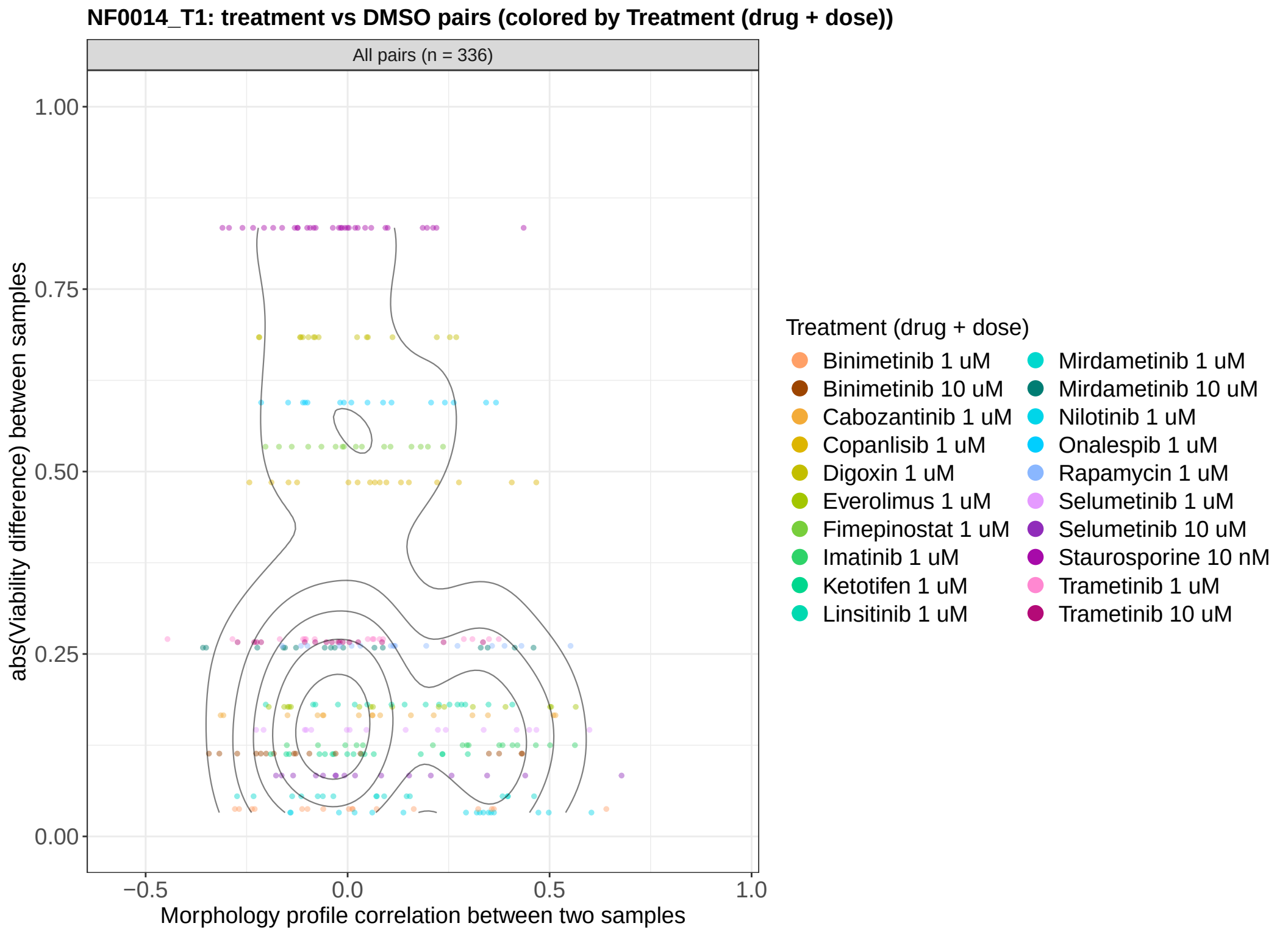
1.00
0.75
0.50
0.25
0.00

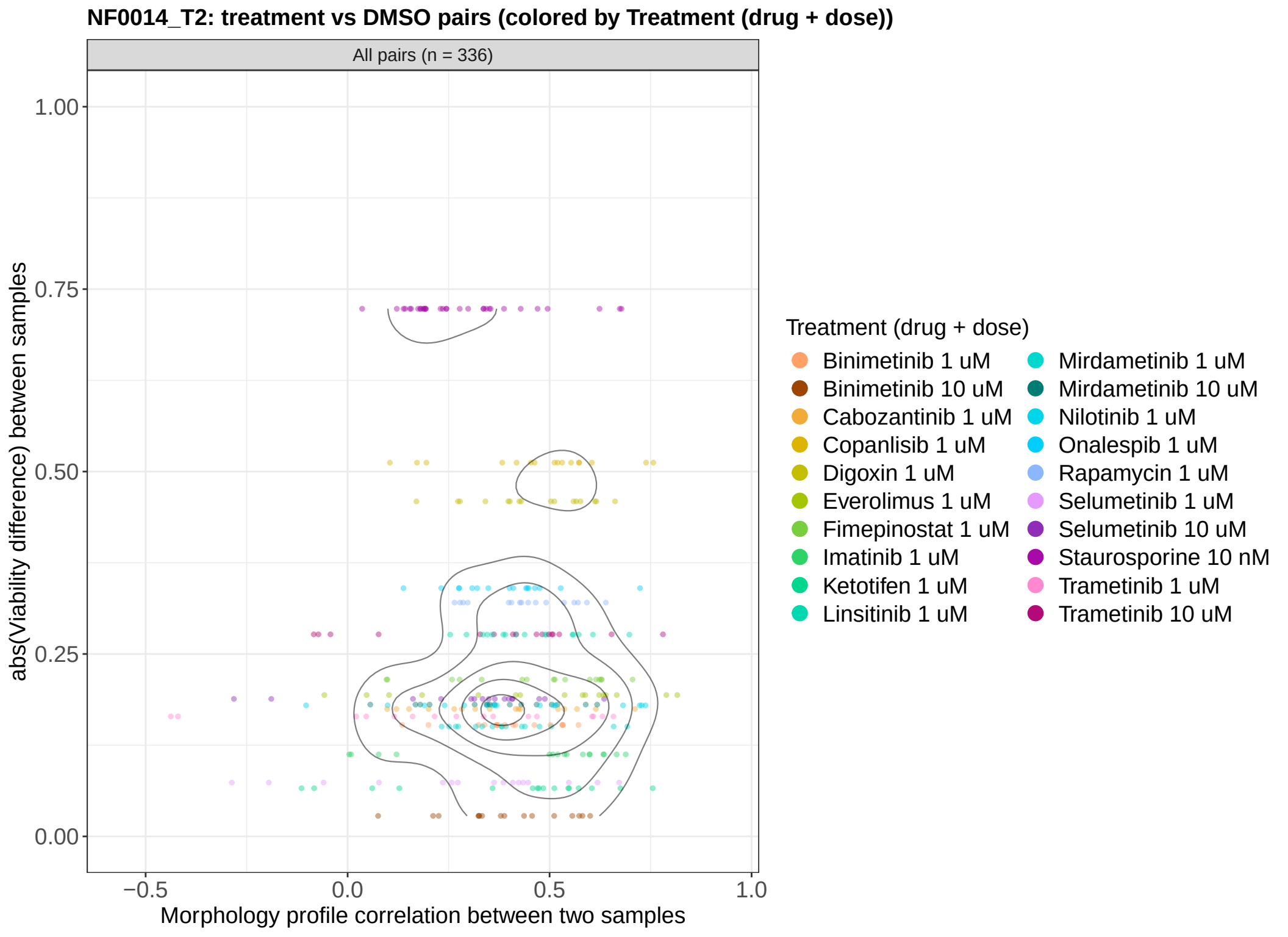
−0.5
0.0
0.5
1.0

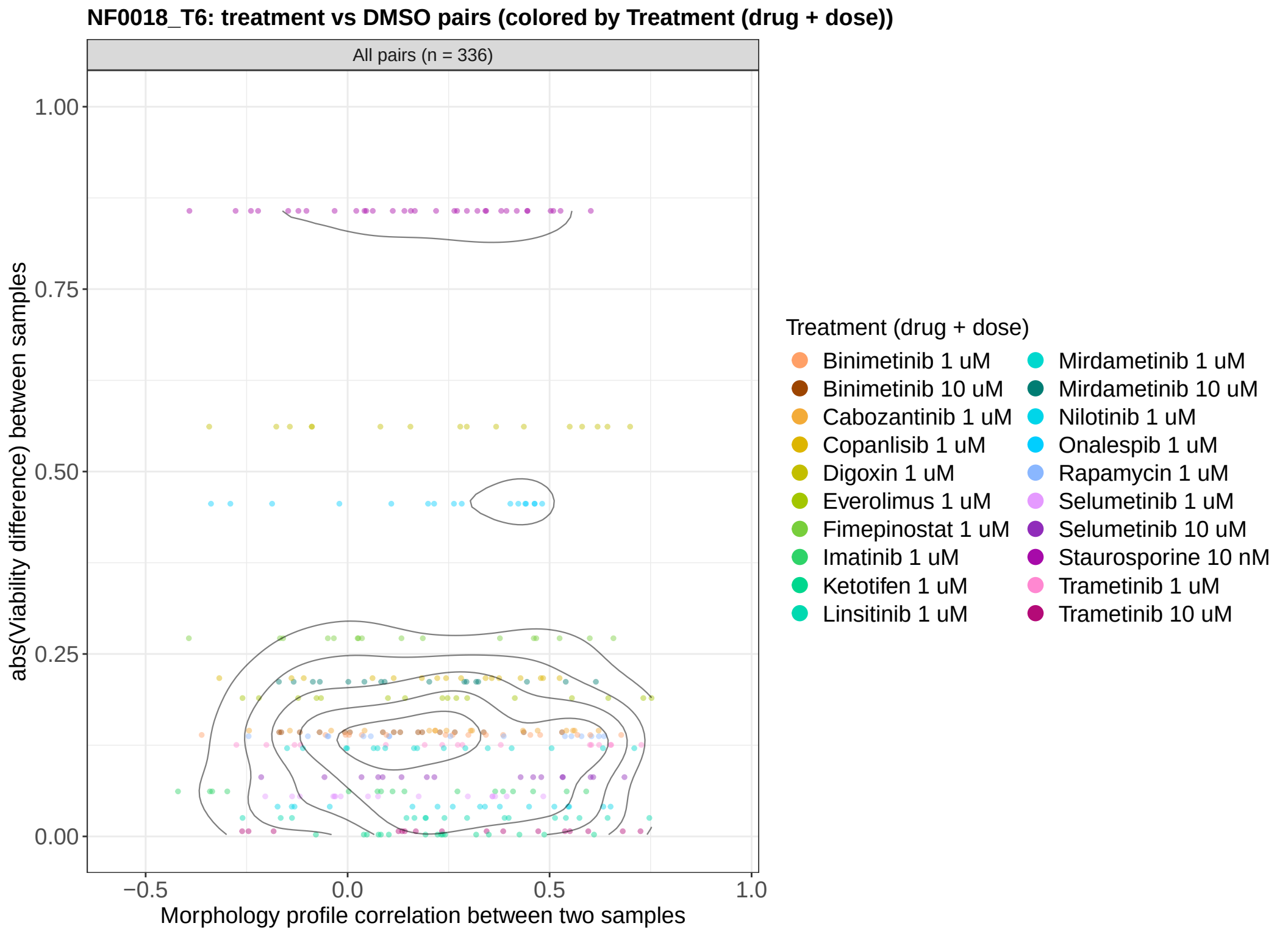
Morphology profile correlation between two samples

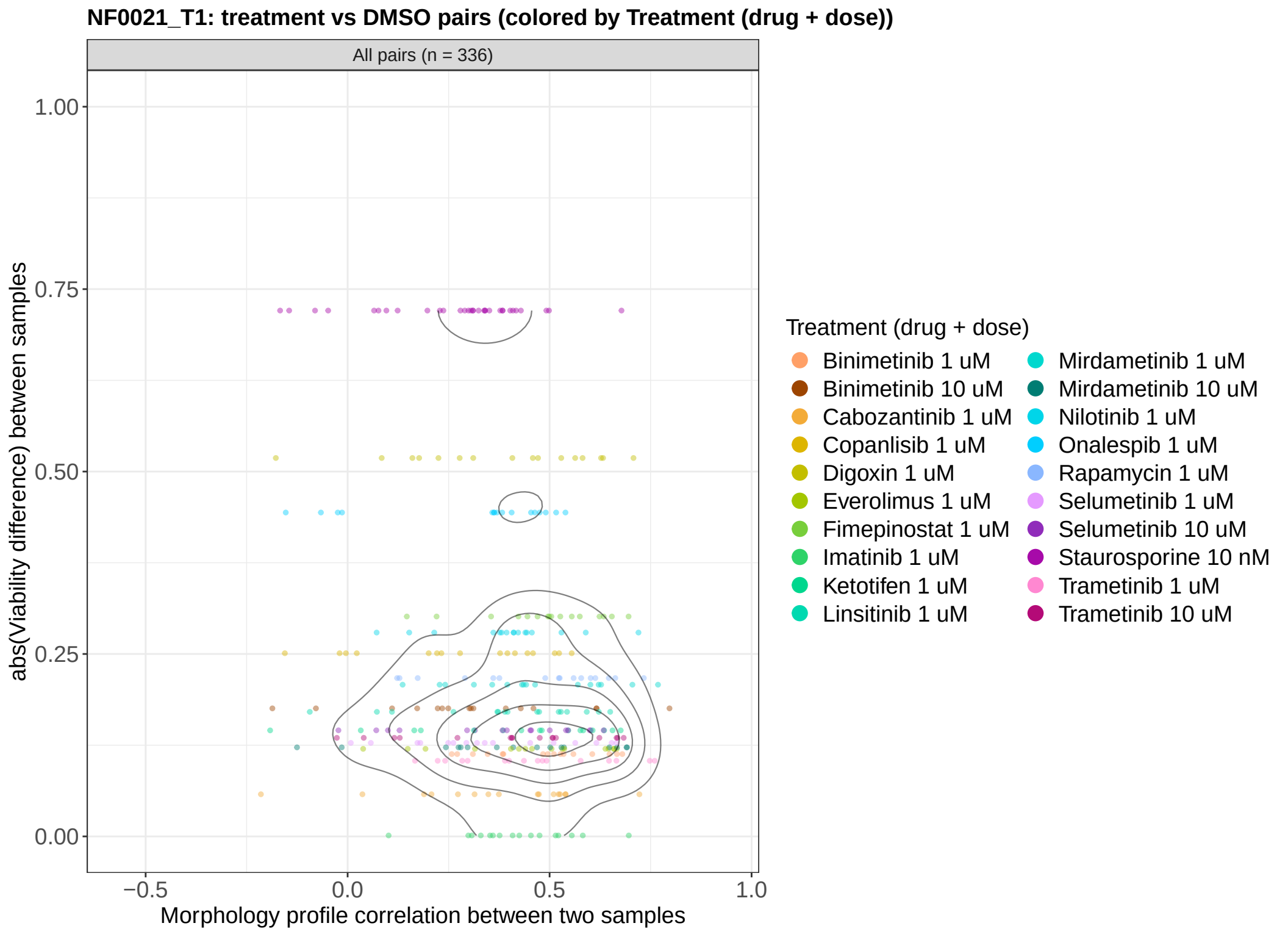

Within−patient pairs (all groups, per−group contours)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
NF0014_T1
NF0014_T2
NF0018_T6
NF0021_T1
NF0035_T1
NF0037_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

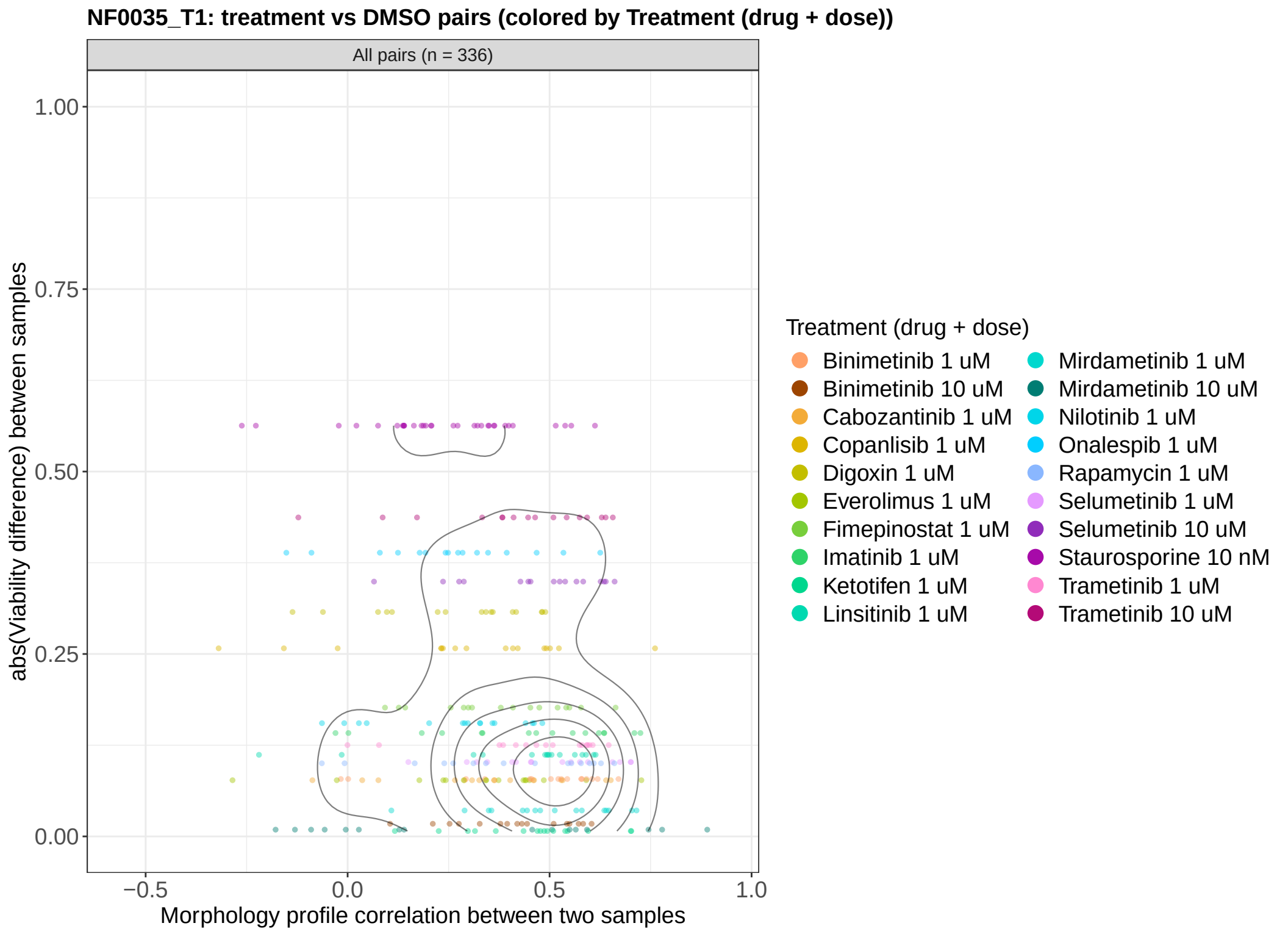

Treatment vs DMSO pairs (all patients) (colored by Treatment (drug + dose))
All pairs (n = 27832)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
ARV−825 1 uM
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Panobinostat 1 uM
Rapamycin 1 uM
Sapanisertib 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trabectedin 1 uM
Trametinib 1 uM
Trametinib 10 uM
Vistusertib 1 uM

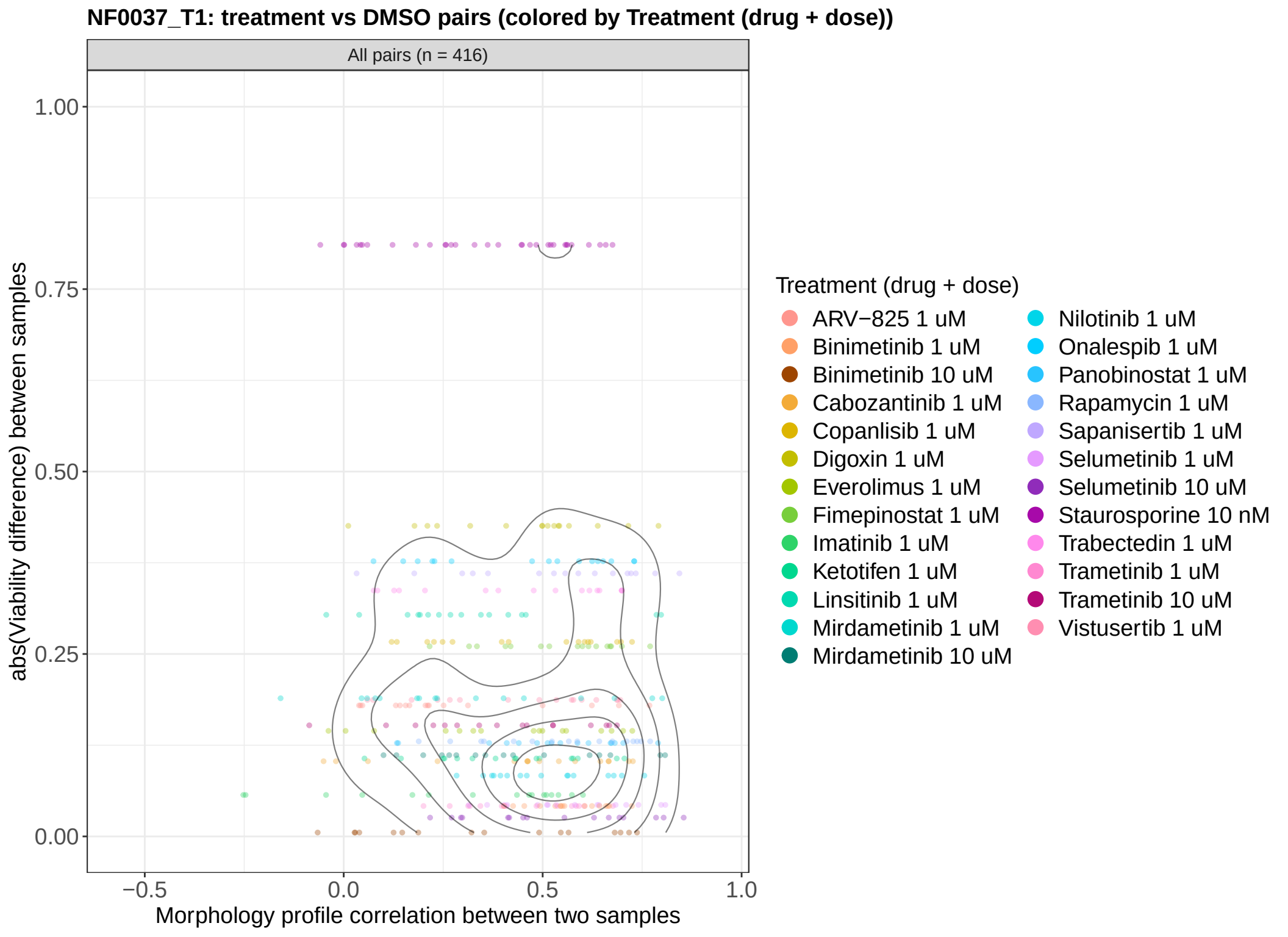

NF0014_T1: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

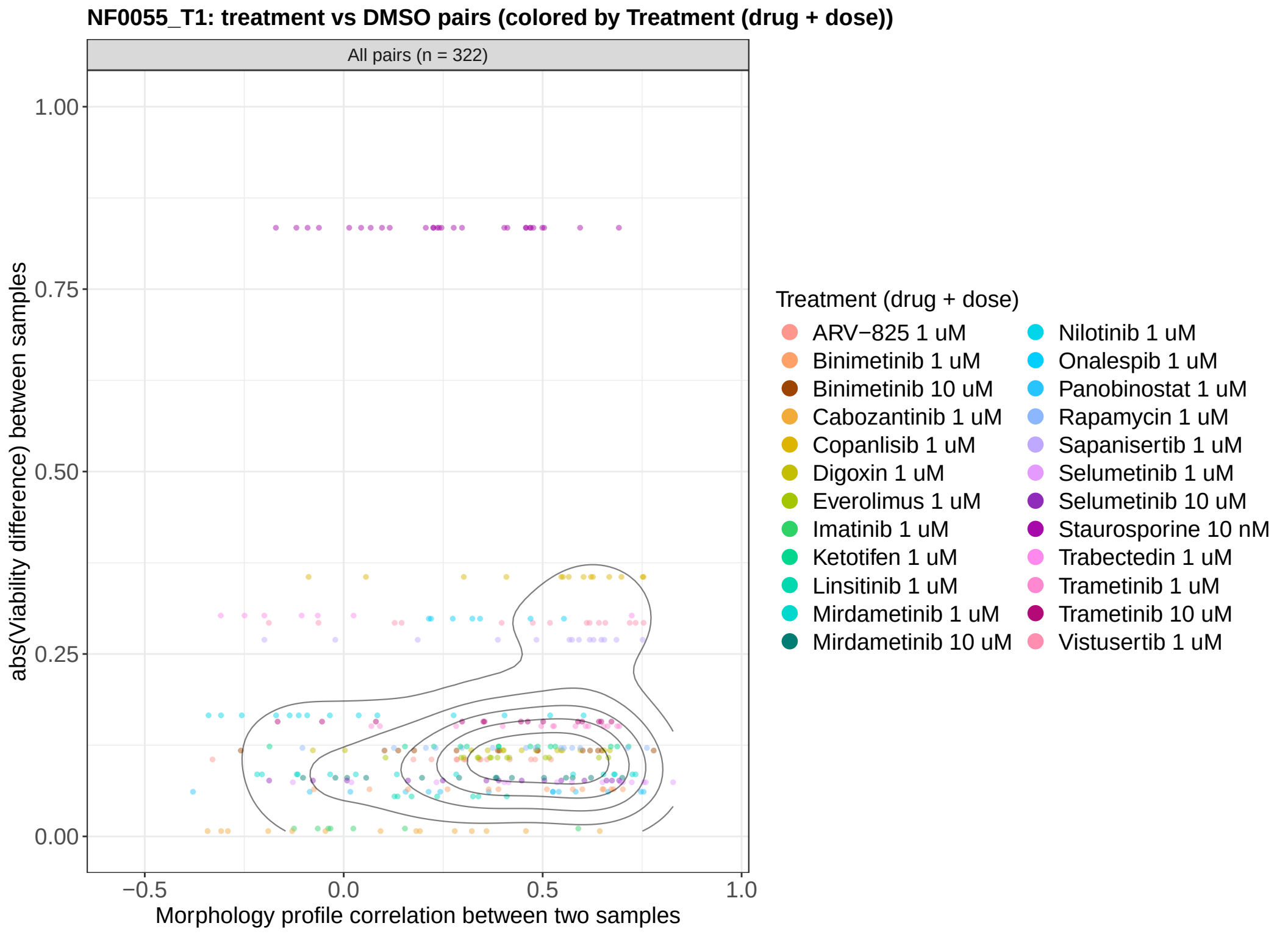

NF0014_T2: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

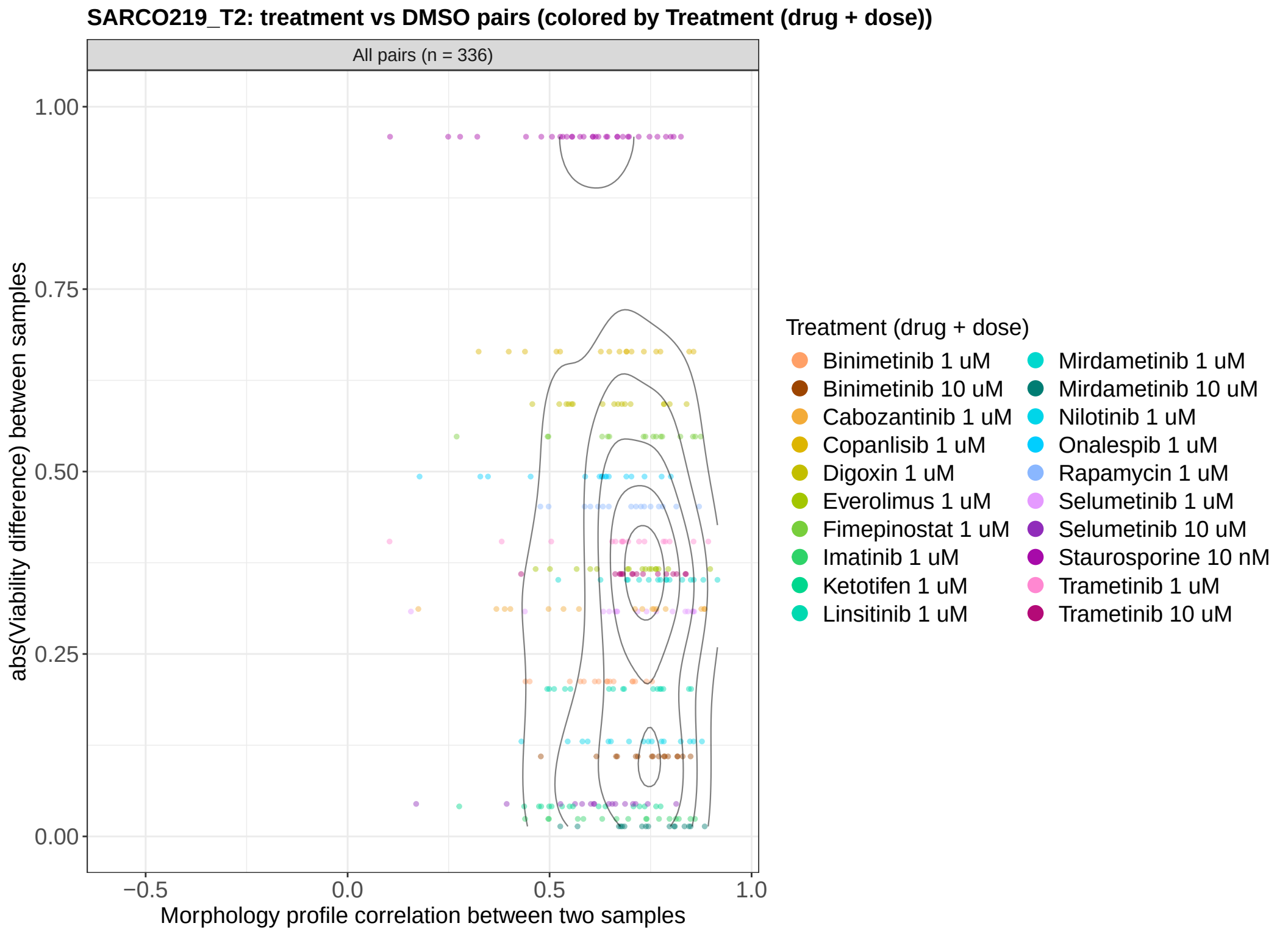

NF0018_T6: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

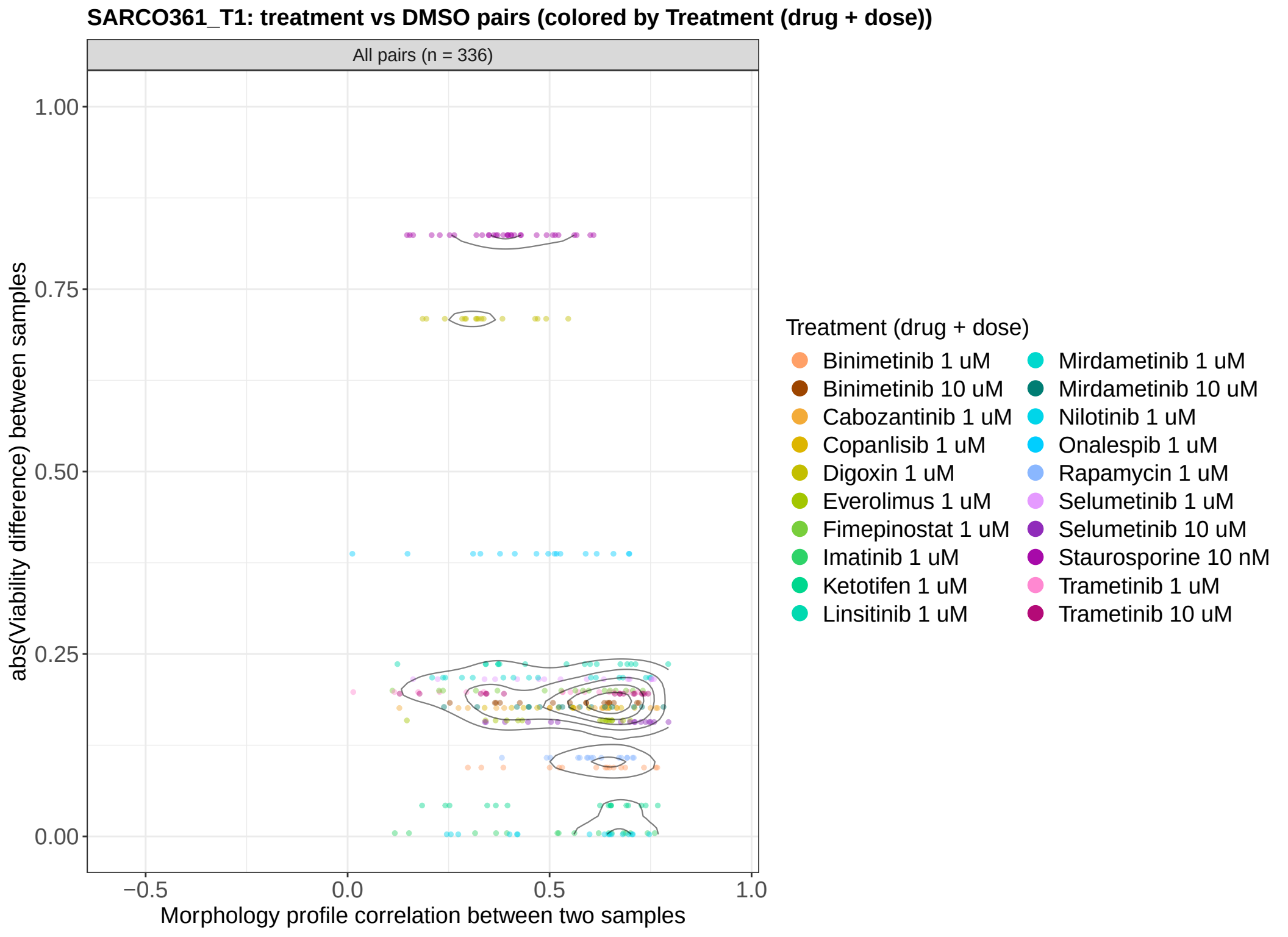

NF0021_T1: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

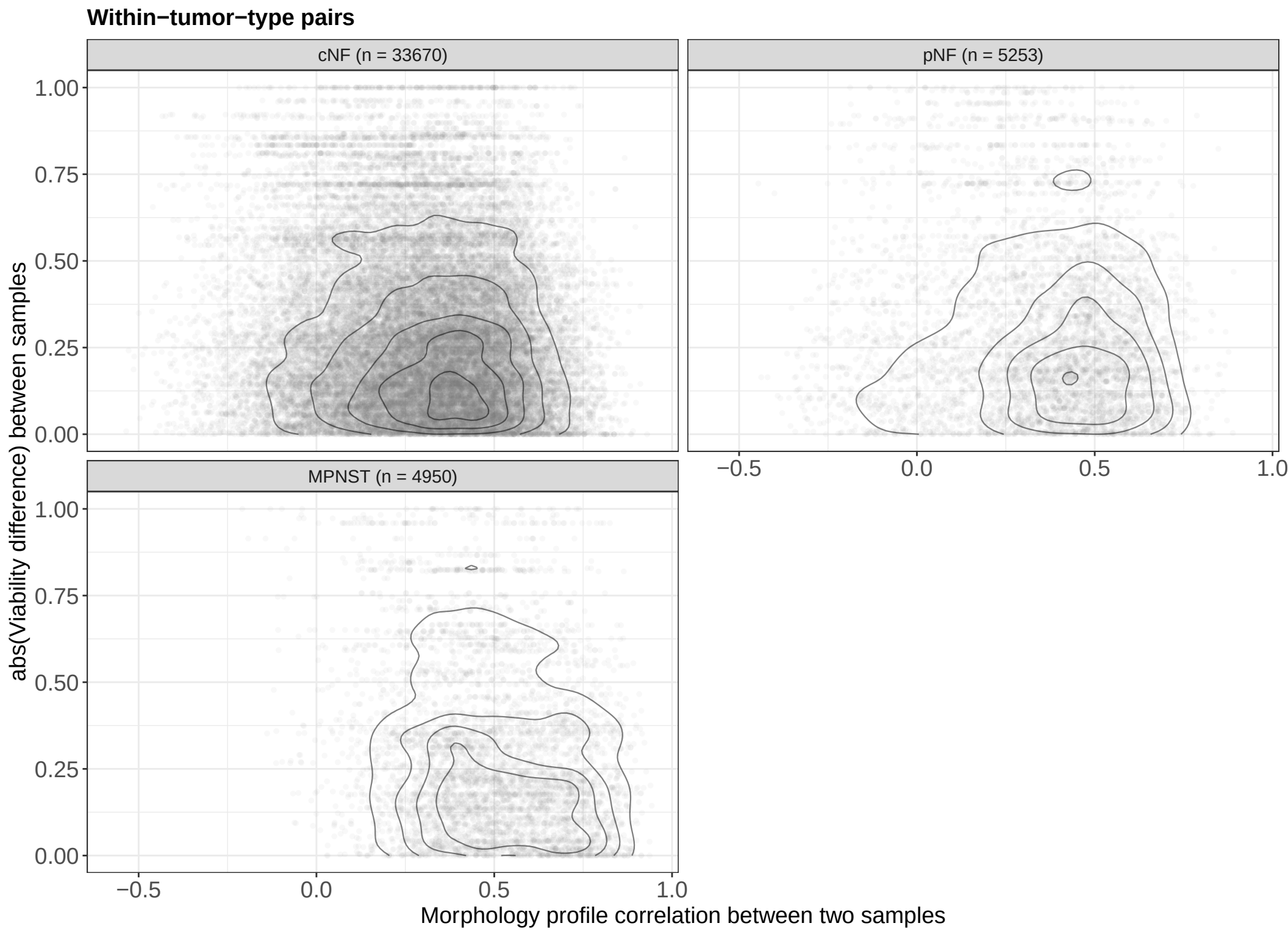

NF0035_T1: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

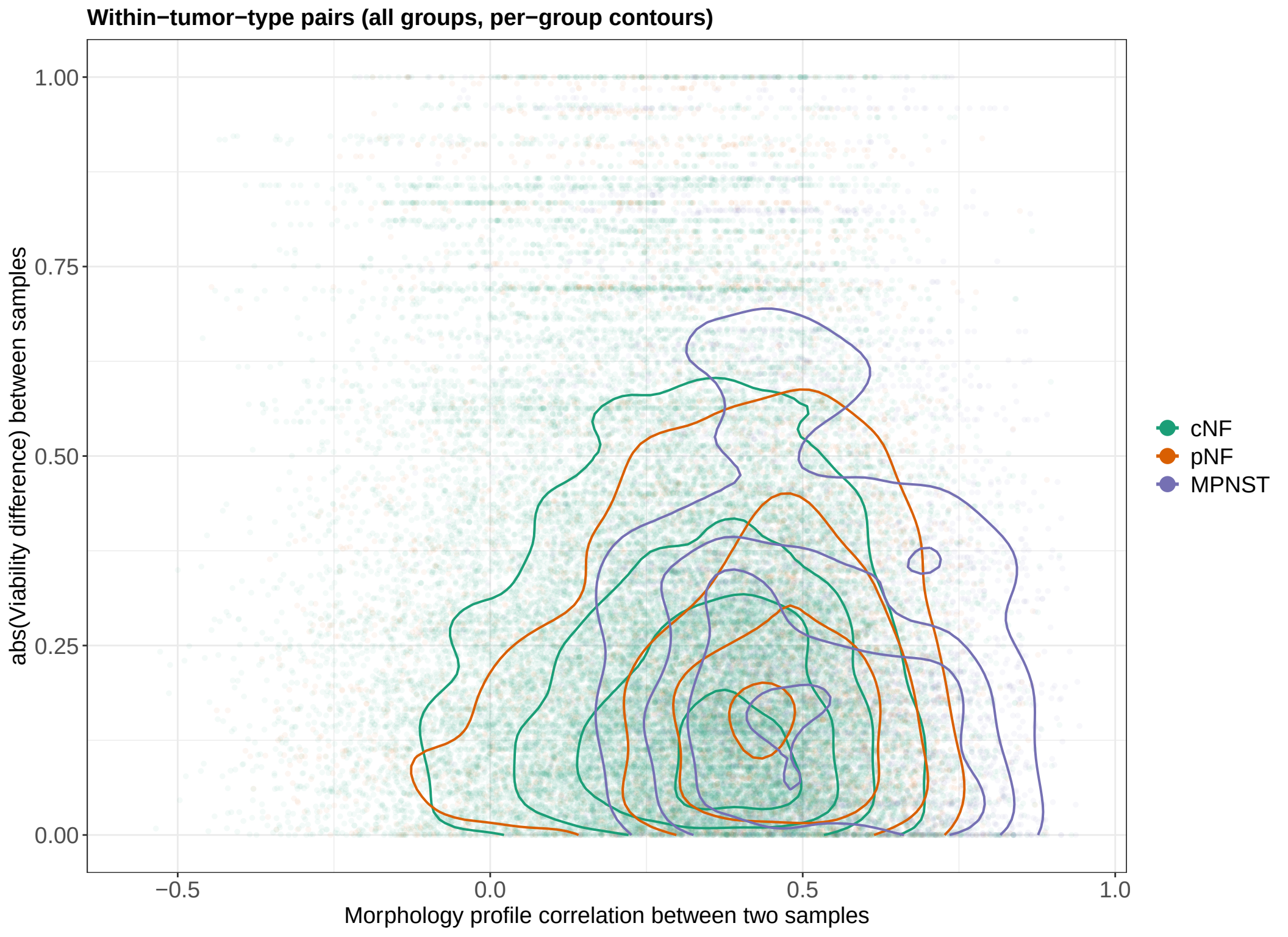

NF0037_T1: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 416)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
ARV−825 1 uM
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Panobinostat 1 uM
Rapamycin 1 uM
Sapanisertib 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trabectedin 1 uM
Trametinib 1 uM
Trametinib 10 uM
Vistusertib 1 uM

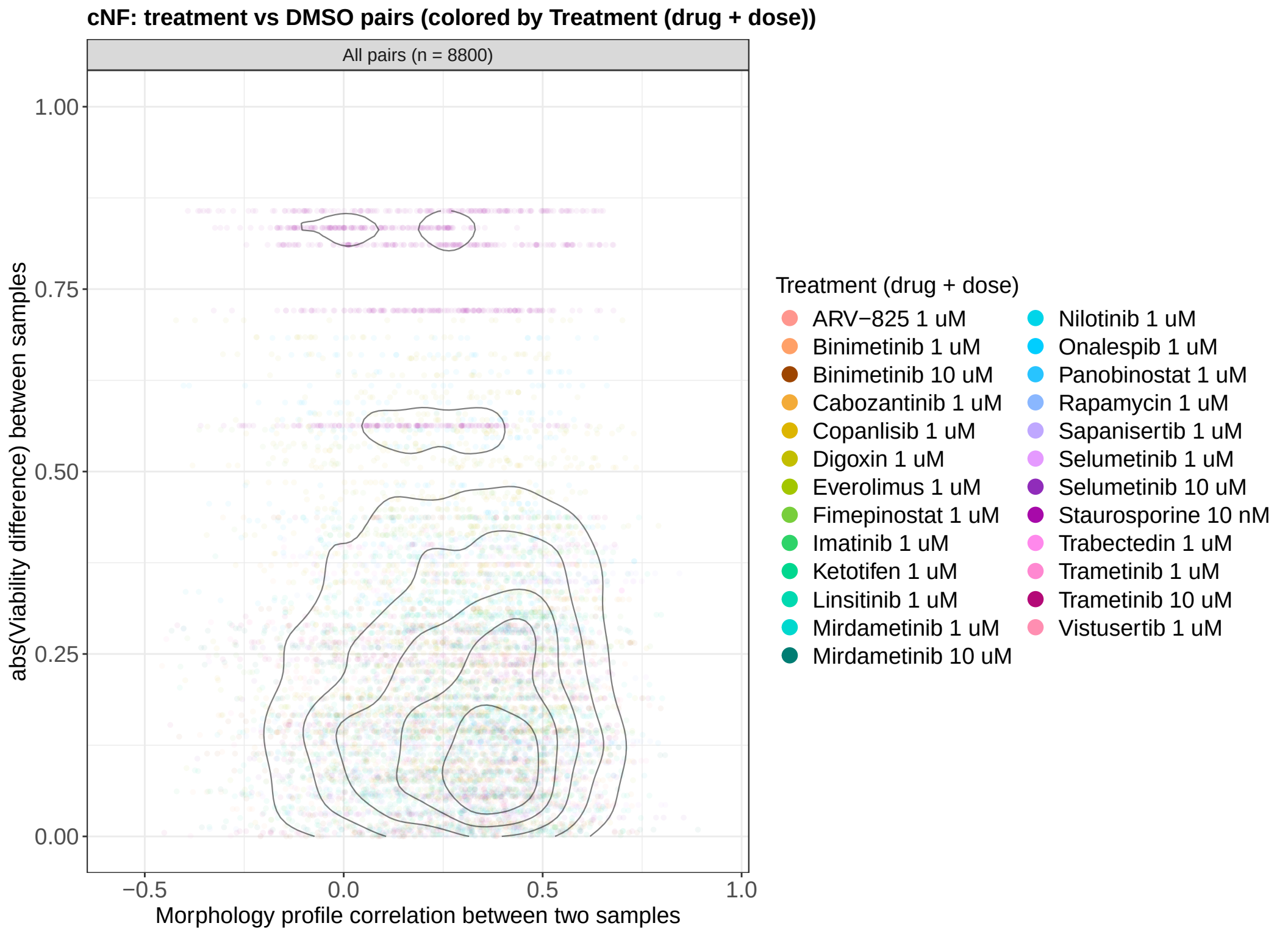

NF0055_T1: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 322)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
ARV−825 1 uM
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Panobinostat 1 uM
Rapamycin 1 uM
Sapanisertib 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trabectedin 1 uM
Trametinib 1 uM
Trametinib 10 uM
Vistusertib 1 uM

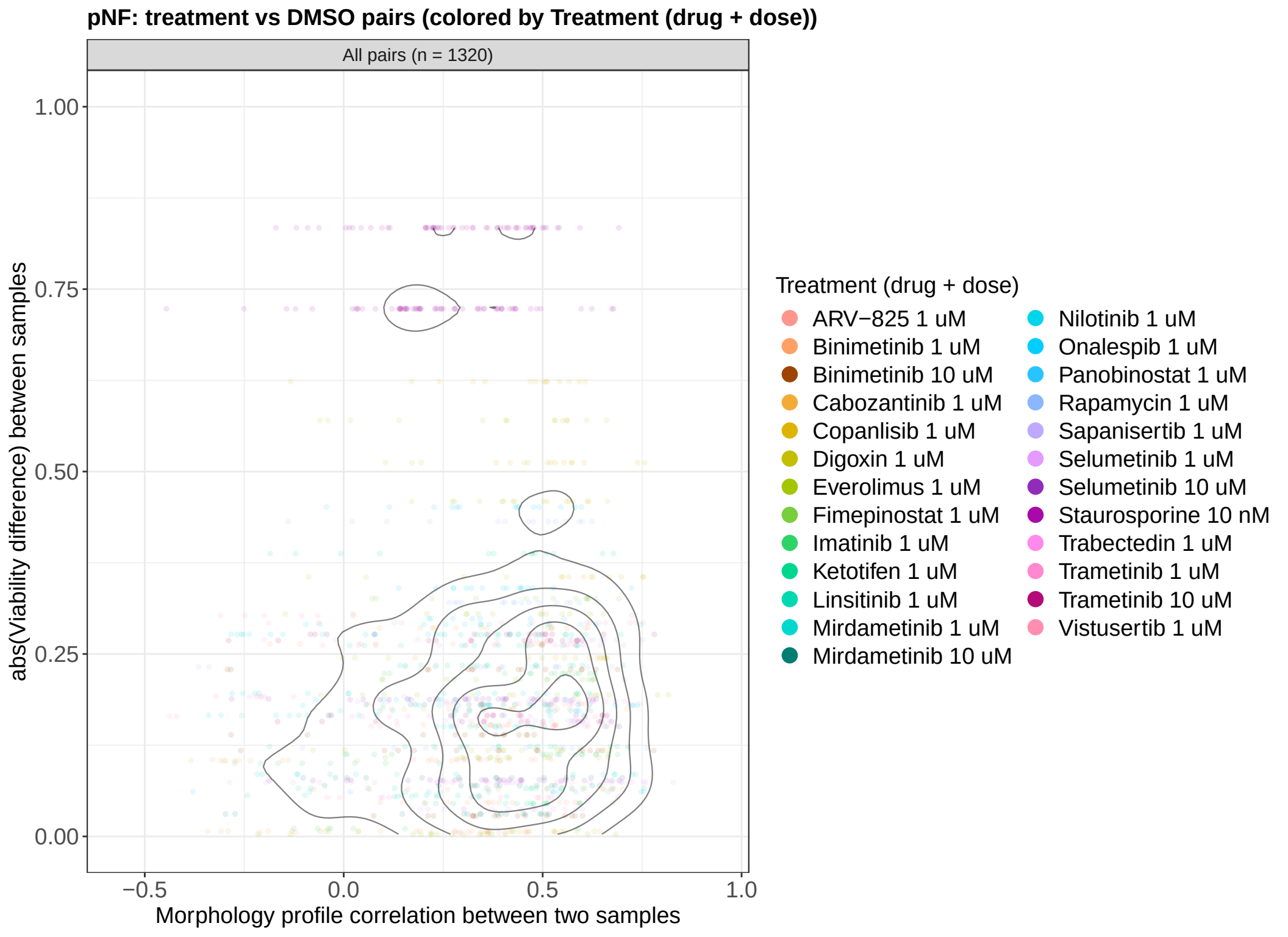

SARCO219_T2: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

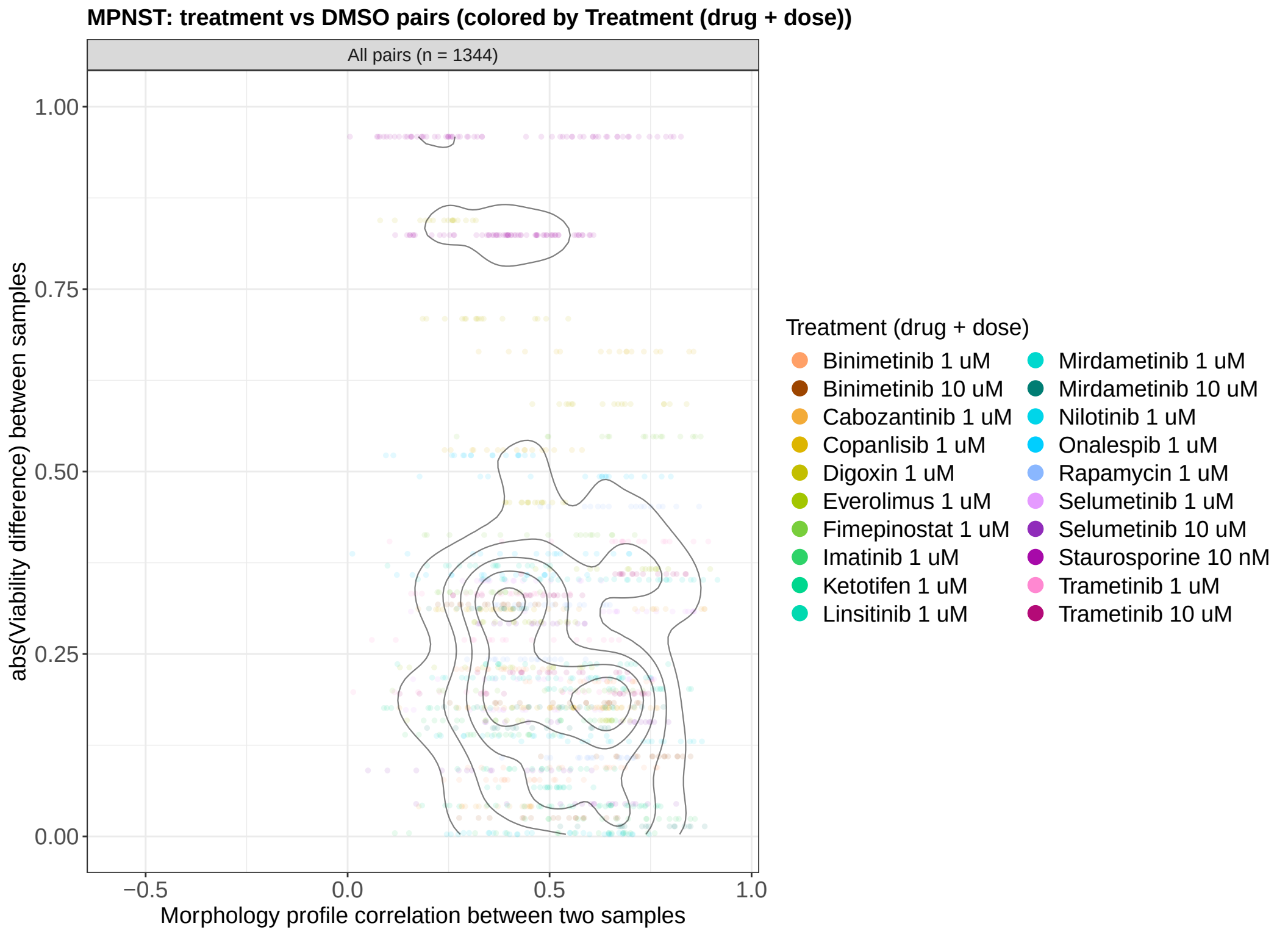

SARCO361_T1: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 336)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM


Within−tumor−type pairs
cNF (n = 33670)
pNF (n = 5253)
MPNST (n = 4950)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples


Within−tumor−type pairs (all groups, per−group contours)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
cNF
pNF
MPNST


cNF: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 8800)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
ARV−825 1 uM
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Panobinostat 1 uM
Rapamycin 1 uM
Sapanisertib 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trabectedin 1 uM
Trametinib 1 uM
Trametinib 10 uM
Vistusertib 1 uM


pNF: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 1320)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
ARV−825 1 uM
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Panobinostat 1 uM
Rapamycin 1 uM
Sapanisertib 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trabectedin 1 uM
Trametinib 1 uM
Trametinib 10 uM
Vistusertib 1 uM


MPNST: treatment vs DMSO pairs (colored by Treatment (drug + dose))
All pairs (n = 1344)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Rapamycin 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trametinib 1 uM
Trametinib 10 uM

Treatment vs DMSO pairs, grouped by drug target

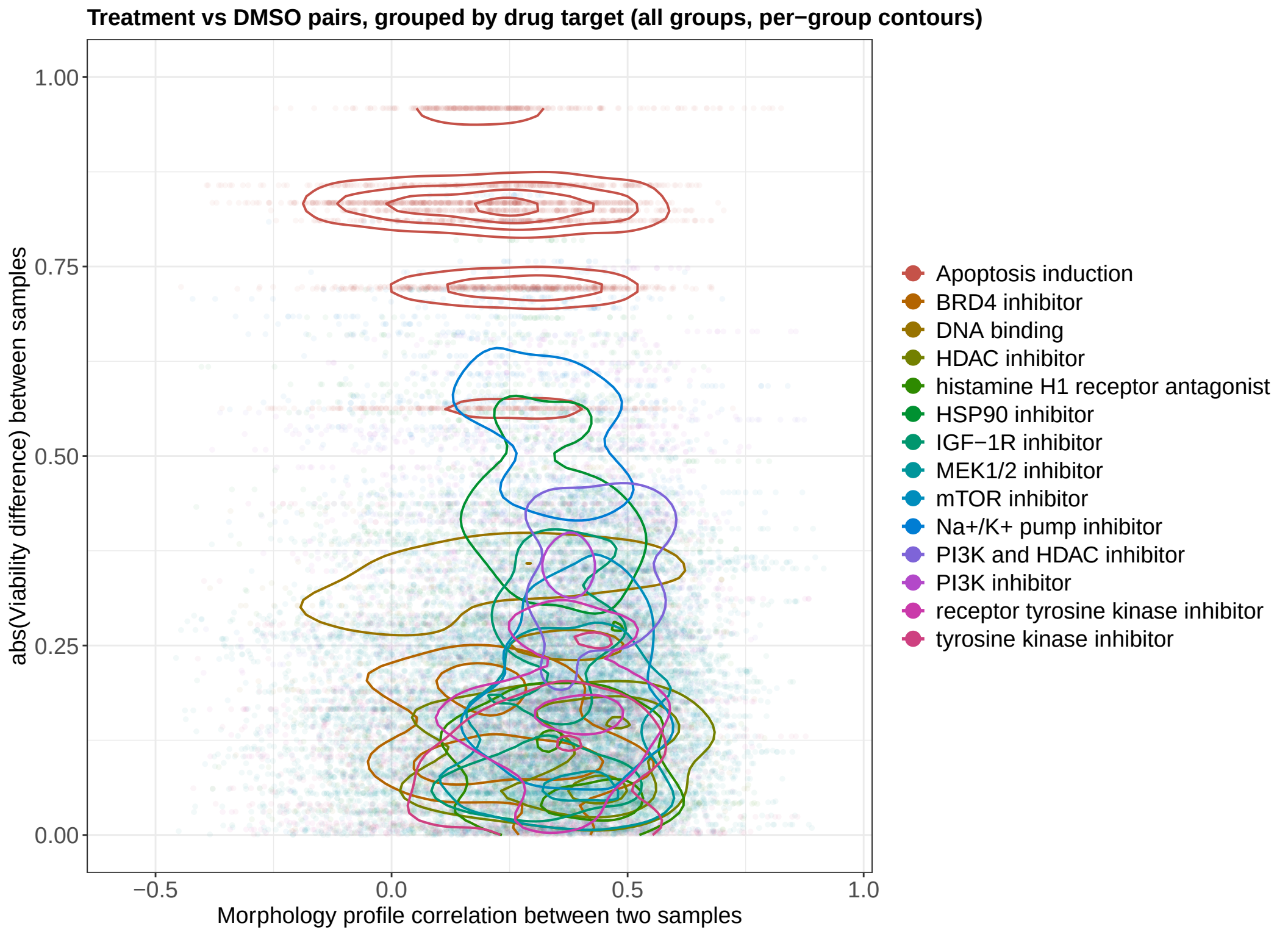
Apoptosis induction (n = 2556)
BRD4 inhibitor (n = 284)
DNA binding (n = 213)
HDAC inhibitor (n = 284)
histamine H1 receptor antagonist (n = 1278)
HSP90 inhibitor (n = 1207)
IGF−1R inhibitor (n = 1207)
MEK1/2 inhibitor (n = 10224)
mTOR inhibitor (n = 3124)
Na+/K+ pump inhibitor (n = 1278)
PI3K and HDAC inhibitor (n = 1136)
PI3K inhibitor (n = 1278)
receptor tyrosine kinase inhibitor (n = 1278)
tyrosine kinase inhibitor (n = 2485)

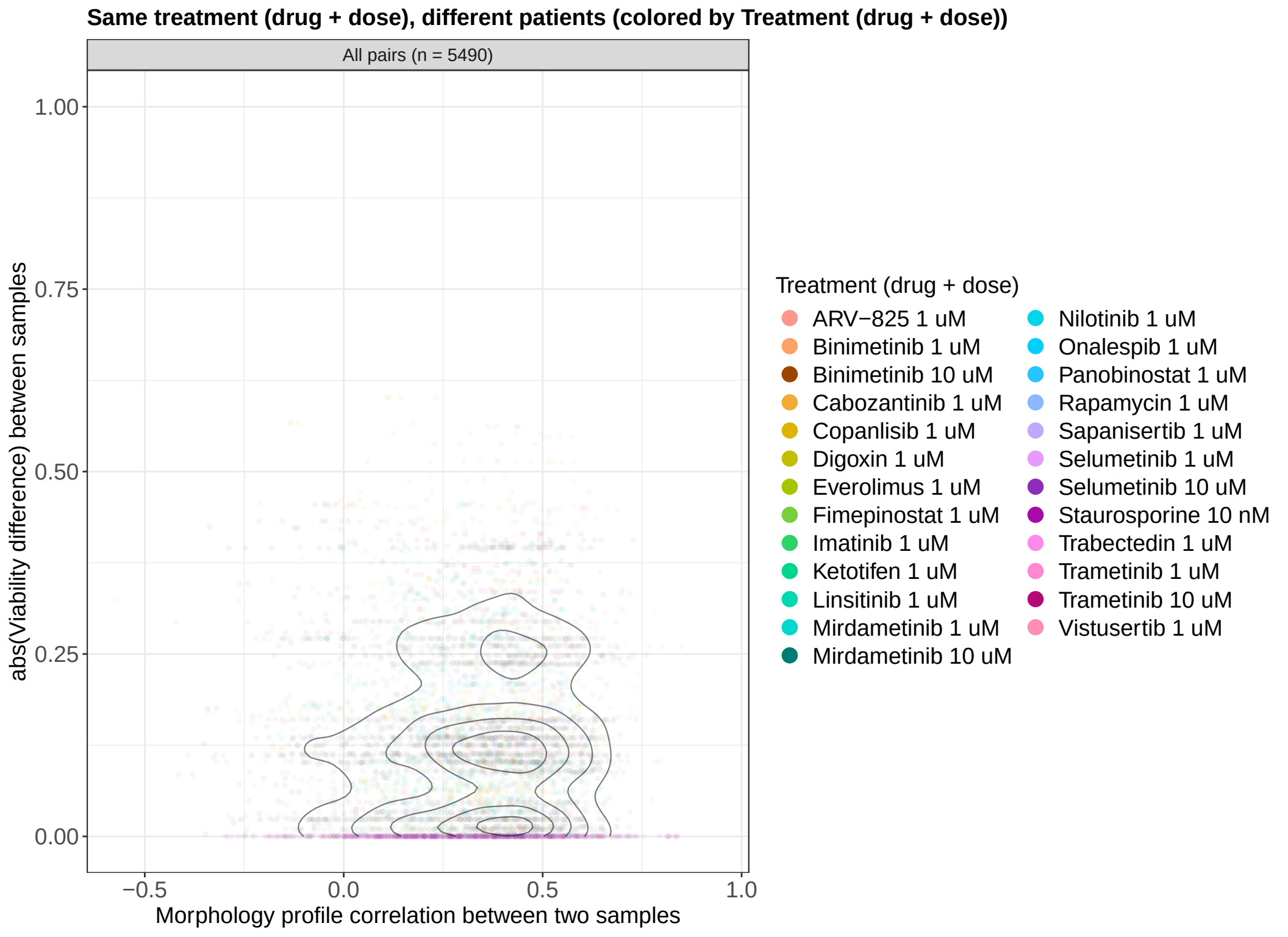
abs(Viability difference) between samples

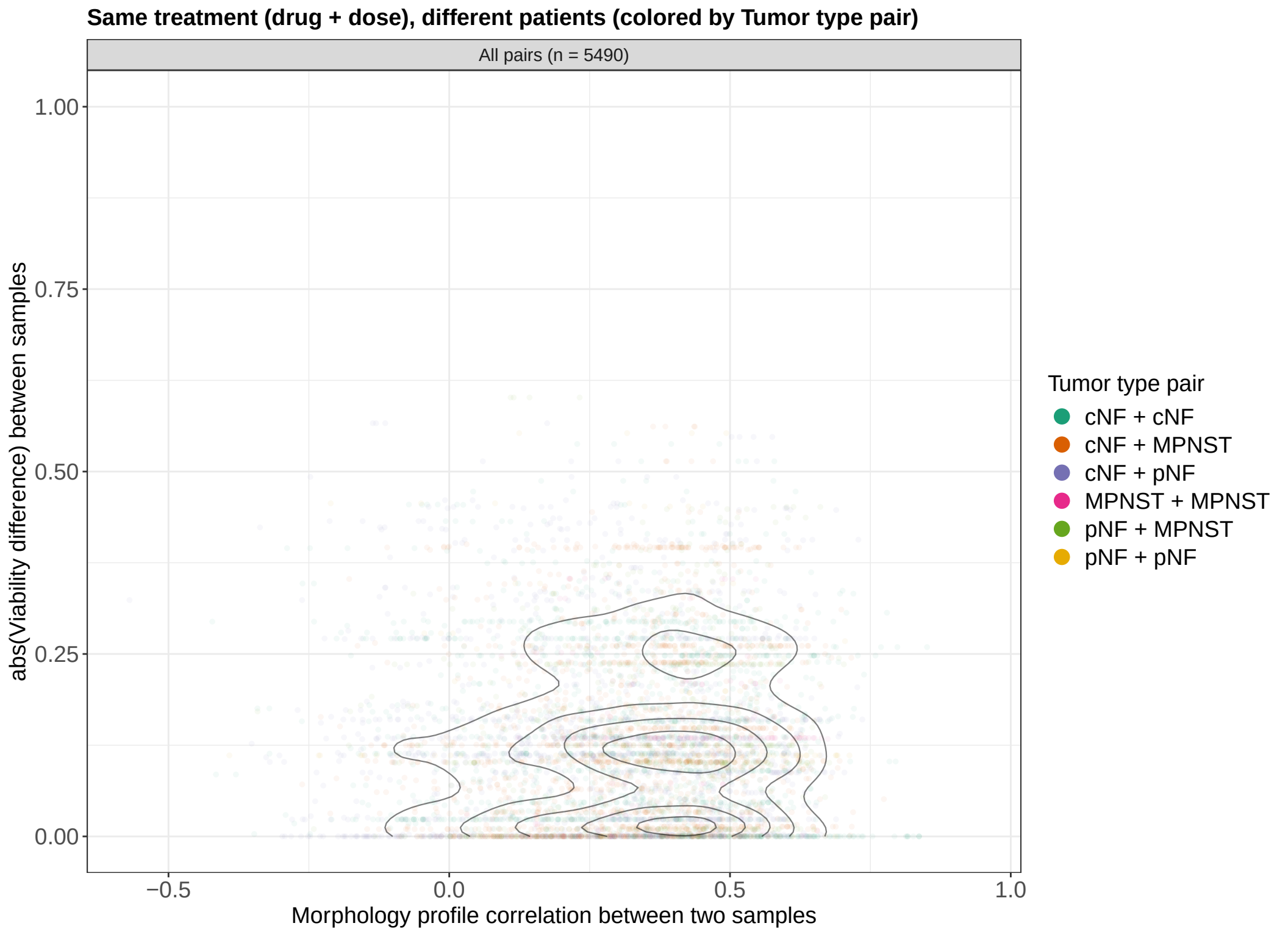
1.00
0.75
0.50
0.25
0.00

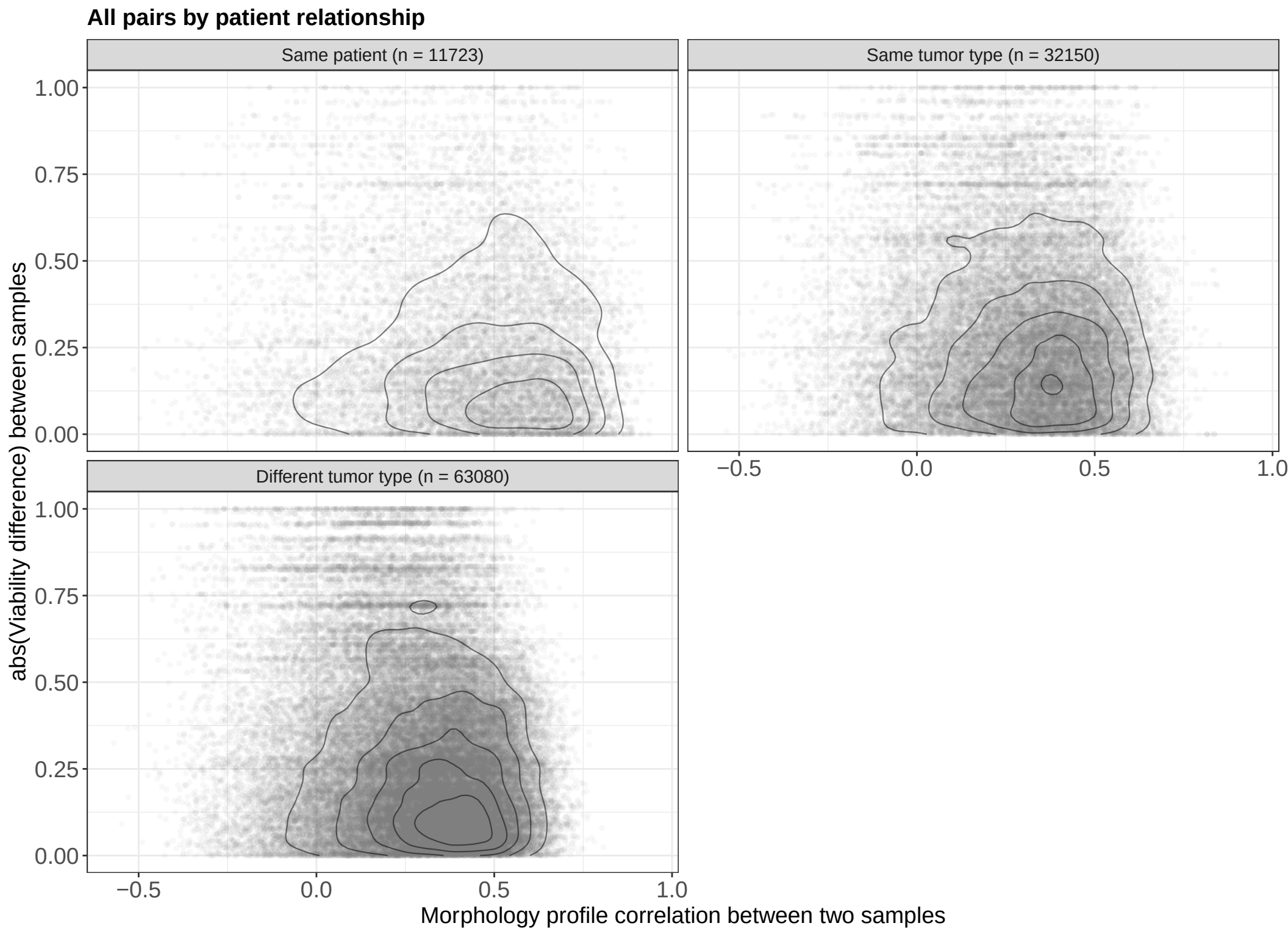
−0.5
0.0
0.5
1.0

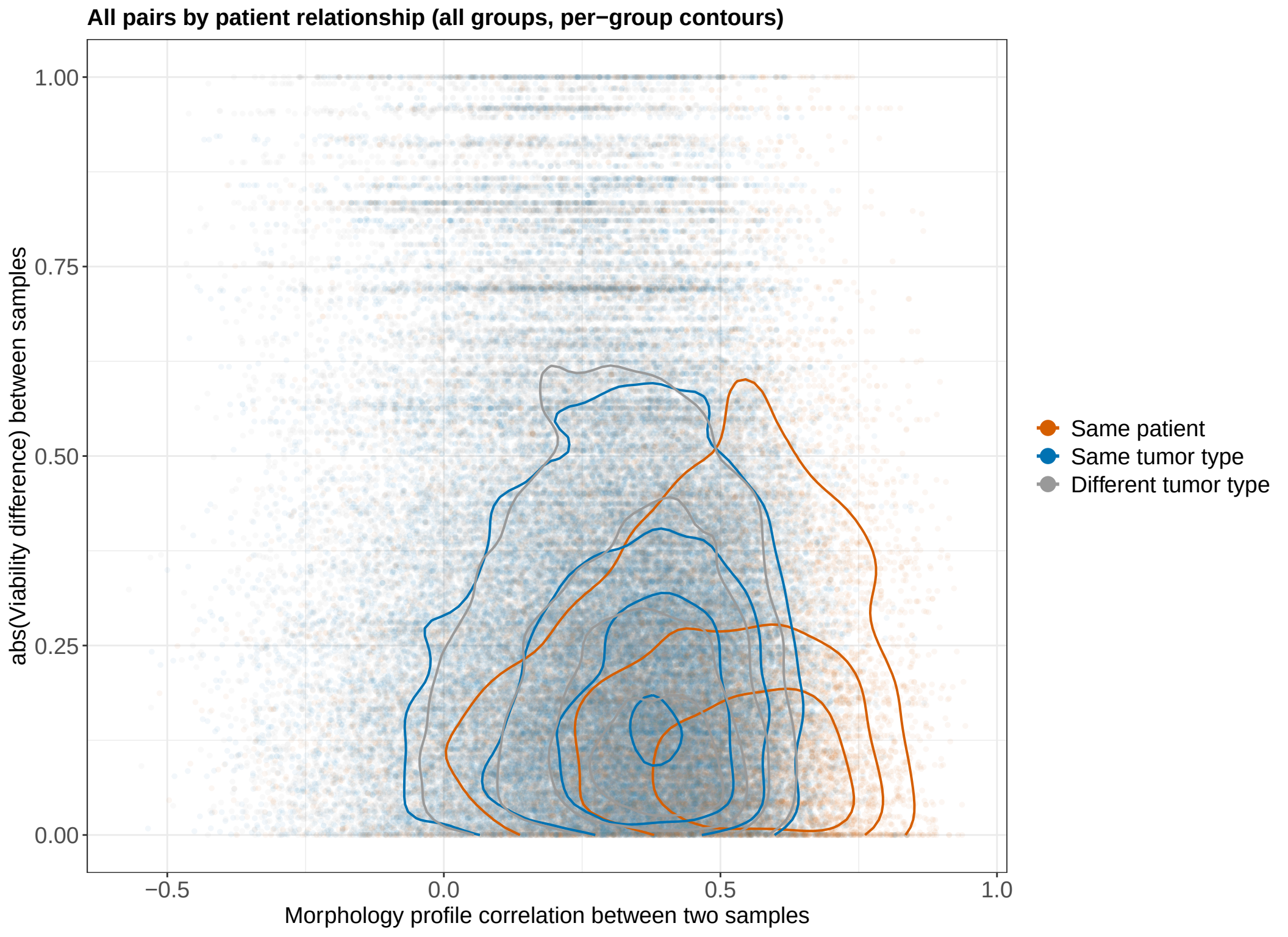
Morphology profile correlation between two samples


Treatment vs DMSO pairs, grouped by drug target (all groups, per−group contours)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Apoptosis induction
BRD4 inhibitor
DNA binding
HDAC inhibitor
histamine H1 receptor antagonist
HSP90 inhibitor
IGF−1R inhibitor
MEK1/2 inhibitor
mTOR inhibitor
Na+/K+ pump inhibitor
PI3K and HDAC inhibitor
PI3K inhibitor
receptor tyrosine kinase inhibitor
tyrosine kinase inhibitor


Same treatment (drug + dose), different patients (colored by Treatment (drug + dose))
All pairs (n = 5490)
1.00
0.75
0.50
0.25
0.00
abs(Viability difference) between samples
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Treatment (drug + dose)
ARV−825 1 uM
Binimetinib 1 uM
Binimetinib 10 uM
Cabozantinib 1 uM
Copanlisib 1 uM
Digoxin 1 uM
Everolimus 1 uM
Fimepinostat 1 uM
Imatinib 1 uM
Ketotifen 1 uM
Linsitinib 1 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Nilotinib 1 uM
Onalespib 1 uM
Panobinostat 1 uM
Rapamycin 1 uM
Sapanisertib 1 uM
Selumetinib 1 uM
Selumetinib 10 uM
Staurosporine 10 nM
Trabectedin 1 uM
Trametinib 1 uM
Trametinib 10 uM
Vistusertib 1 uM


Same treatment (drug + dose), different patients (colored by Tumor type pair)
All pairs (n = 5490)
1.00
0.75
0.50
0.25
0.00
abs(Viability difference) between samples
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Tumor type pair
cNF + cNF
cNF + MPNST
cNF + pNF
MPNST + MPNST
pNF + MPNST
pNF + pNF


All pairs by patient relationship
Same patient (n = 11723)
Same tumor type (n = 32150)
Different tumor type (n = 63080)
abs(Viability difference) between samples
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples


All pairs by patient relationship (all groups, per−group contours)
1.00
0.75
0.50
0.25
0.00
abs(Viability difference) between samples
−0.5
0.0
0.5
1.0
Morphology profile correlation between two samples
Same patient
Same tumor type
Different tumor type